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Q1.(Creating Sets)

```
integer_set = {10, 20, 30, 40, 50}
mixed_set = {100, "Python", 3.14}
empty_set = set()
string_set = set("hello")
print("a) Integer Set:", integer_set)
print("Type:", type(integer_set))
print("\nb) Mixed Data
```

Q2.(Characteristics of Sets)

```
numbers = {10, 20, 30, 20, 40, 10, 50}
print("Set:", numbers)
print("\nPrinting the set multiple times:")
print(numbers)
print(numbers)
print(numbers)
```

Q3.(Membership Testing)

```
colors = set()
print("Enter 5 colors:")
for i in range(5):
```

```
color = input(f"Enter color {i+1}: ")
colors.add(color)
print("\nColors Set:", colors)
search_color = input("\nEnter a color to search: ")
if search_color in colors:
    print(search_color, "is present in the set.")
if search_color not in colors:
    print(search_color, "is not present in the set.")
```

```
Q4.(add(), remove(), discard())
fruits = {'apple', 'banana', 'mango'}
print("Original Set:", fruits)
fruits.add('orange')
print("\nAfter add('orange'):", fruits)
fruits.add('banana')
print("\nAfter add('banana') again:", fruits)
fruits.remove('mango')
print("\nAfter remove('mango'):", fruits)
fruits.discard('grape')
print("\nAfter discard('grape'):", fruits)
```

```
Q5.(pop() and clear())
s = {100, 200, 300, 400, 500}
```

```
print("Original Set:", s)
removed_element = s.pop()
print("\nRemoved Element:", removed_element)
print("Set after pop():", s)
s.clear()
print("\nFinal Set after clear():", s)
```

Q6.(update() and copy())

```
set1 = {1, 2, 3}
set2 = {3, 4, 5, 6}
set1_copy = set1.copy()
print("Original Copy of set1:", set1_copy)
set1.update(set2)
print("Updated set1:", set1)
print("Original Copy (unchanged):", set1_copy)
```

Q7.(Union and Intersection)

```
A = {1, 2, 3, 4, 5}
B = {4, 5, 6, 7, 8}
print("Union using union():", A.union(B))
print("Union using | operator:", A | B)
print("Intersection using intersection():", A.intersection(B))
print("Intersection using & operator:", A & B)
```

Q8.(Combined Methods)

```
numbers = set()
print("Enter 6 numbers:")
for i in range(6):
    num = int(input(f"Enter number {i+1}: "))
    numbers.add(num)
print("\nSet after input:", numbers)
num1 = int(input("\nEnter first number to add: "))
numbers.add(num1)
num2 = int(input("Enter second number to add: "))
numbers.add(num2)
print("\nSet after adding two numbers:", numbers)
remove_num = int(input("\nEnter a number to remove: "))
numbers.discard(remove_num)
print("\nFinal Set:", numbers)
print("Length of the Set:", len(numbers))
```

Q9.(Practical Application)

```
scores = [85, 92, 78, 92, 85, 88, 95, 78]
print("Original Scores:", scores)
```

```
unique_scores = set(scores)

print("\nUnique Scores (Set):", unique_scores)

sorted_scores = sorted(list(unique_scores))

print("\nSorted Unique Scores:", sorted_scores)

print("\nTotal Unique Scores:", len(unique_scores))
```

Q10.(Mini Project - Combined Concepts)

```
items = set()

print("\n===== UNIQUE ITEM COLLECTOR =====")

print("1. Add Item")

print("2. Remove Item")

print("3. Show All Unique Items")

print("4. Check if an Item Exists")

print("5. Clear All Items")

print("6. Exit")

choice = int(input("Enter your choice (1-6): "))

if choice == 1:

    item = input("Enter item to add: ")

    items.add(item)

    print(item, "added successfully.")

elif choice == 2:

    item = input("Enter item to remove: ")

    items.discard(item)
```

```
    print(item, "removed (if it existed).")
elif choice == 3:
    if len(items) == 0:
        print("No items available.")
    else:
        print("Unique Items:", items)
elif choice == 4:
    item = input("Enter item to check: ")
    if item in items:
        print(item, "is present in the set.")
    else:
        print(item, "is not present in the set.")
elif choice == 5:
    items.clear
```